Group 2:
Anastasia
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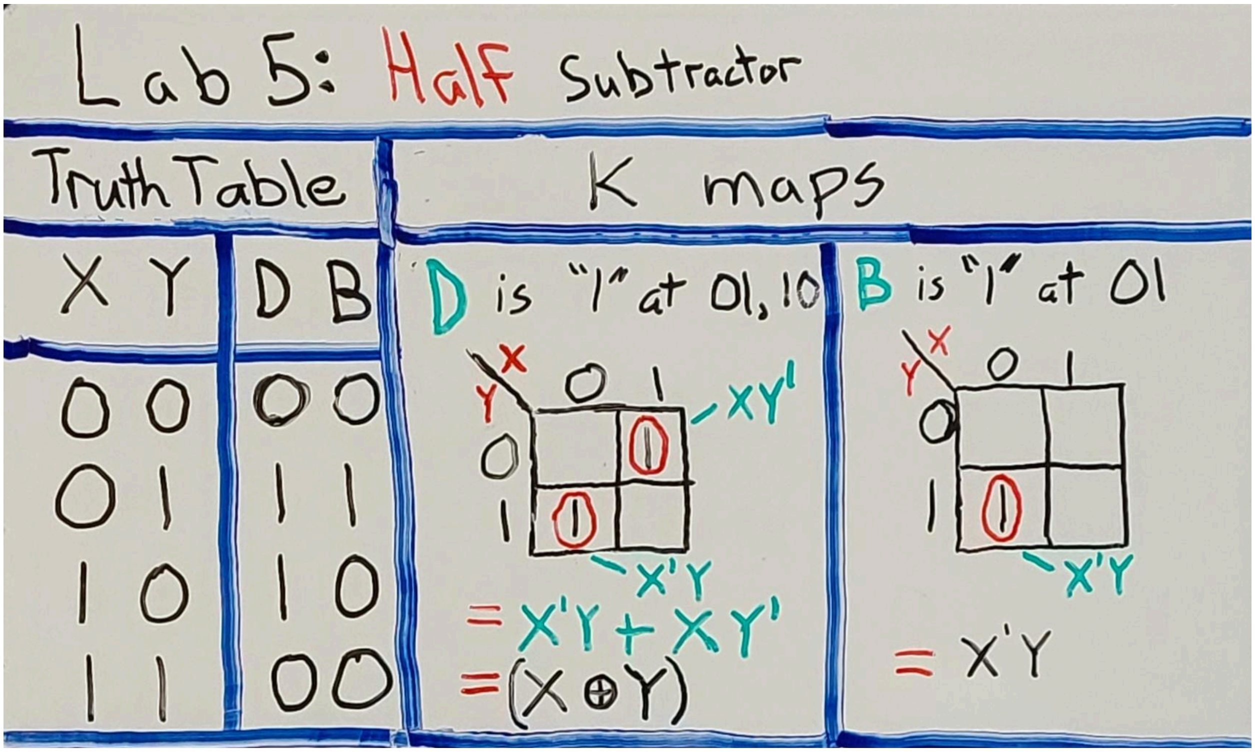
Moises

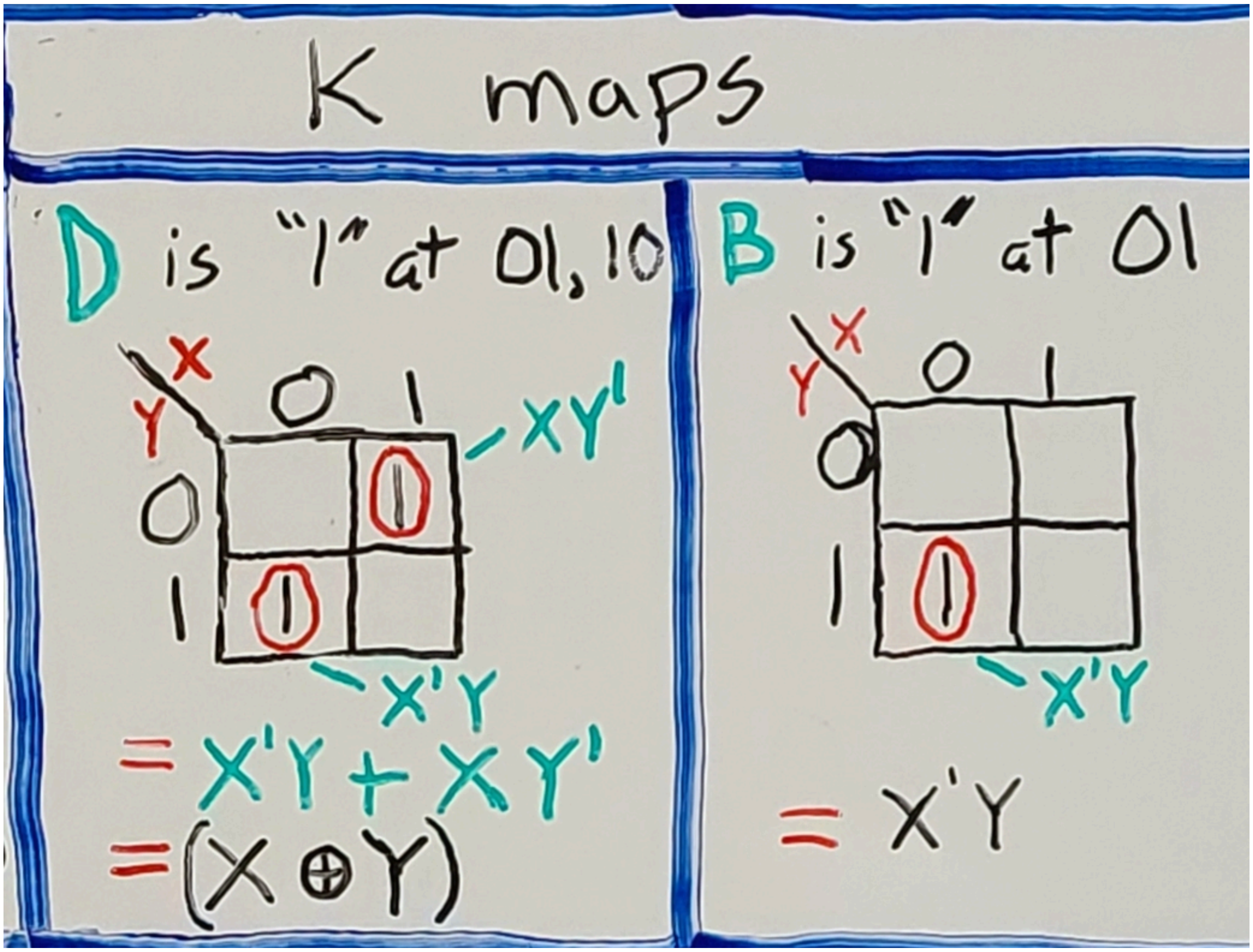
Lab Report #5

Section 5.1

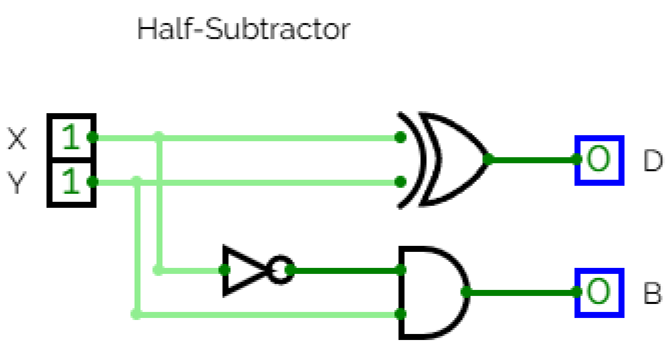
1) Truth Table



2) B and D in terms of x and y variables



3) Circuit Diagram



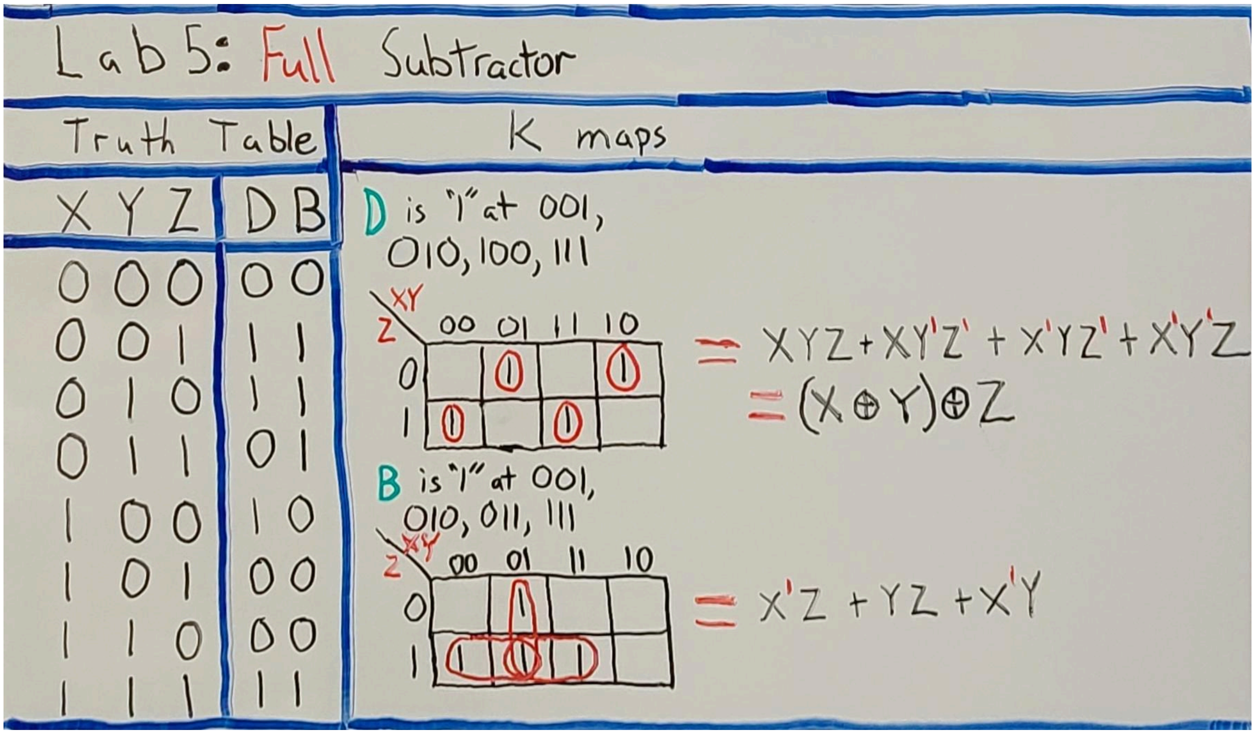
https://circuitverse.org/users/403924/projects/lab-5-group-2

4) Observations

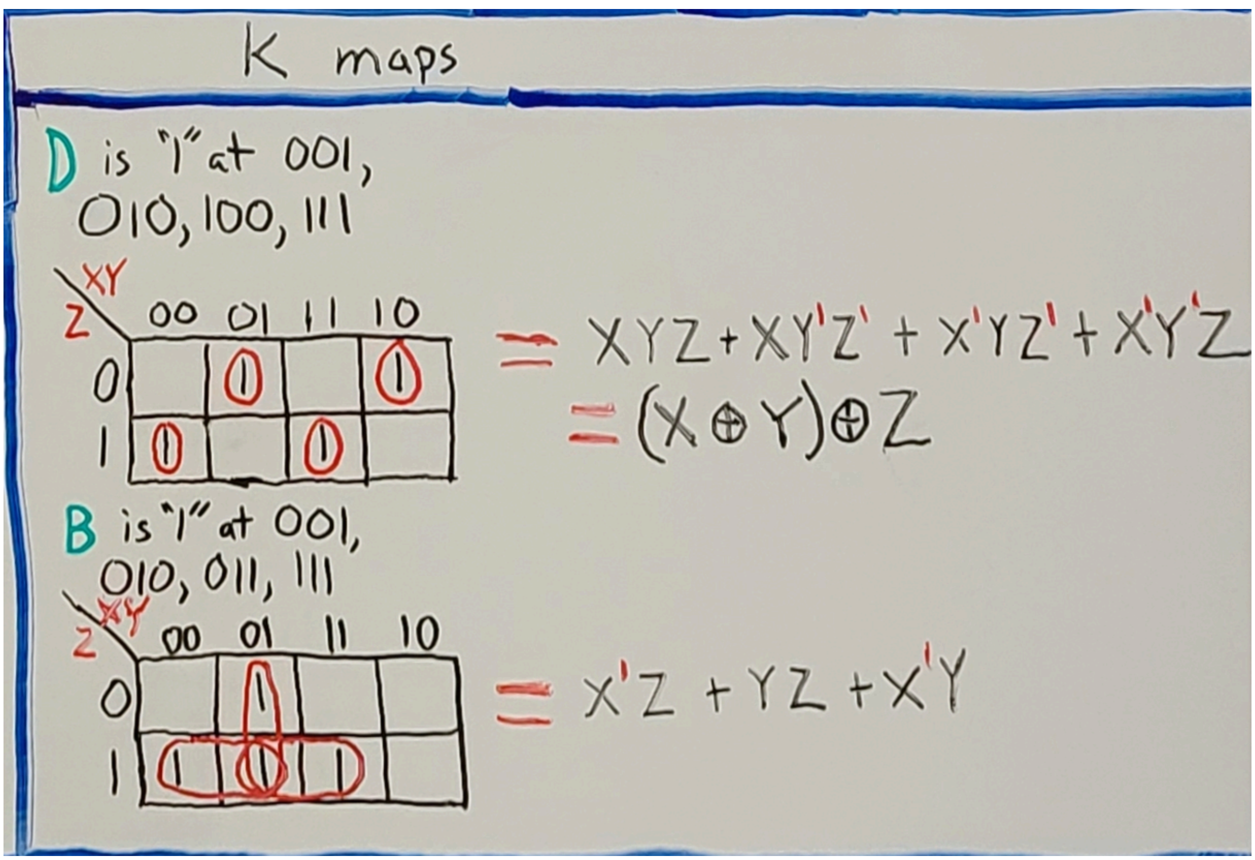
Our circuit was observed to match our expected truth table values.

Section 5.2

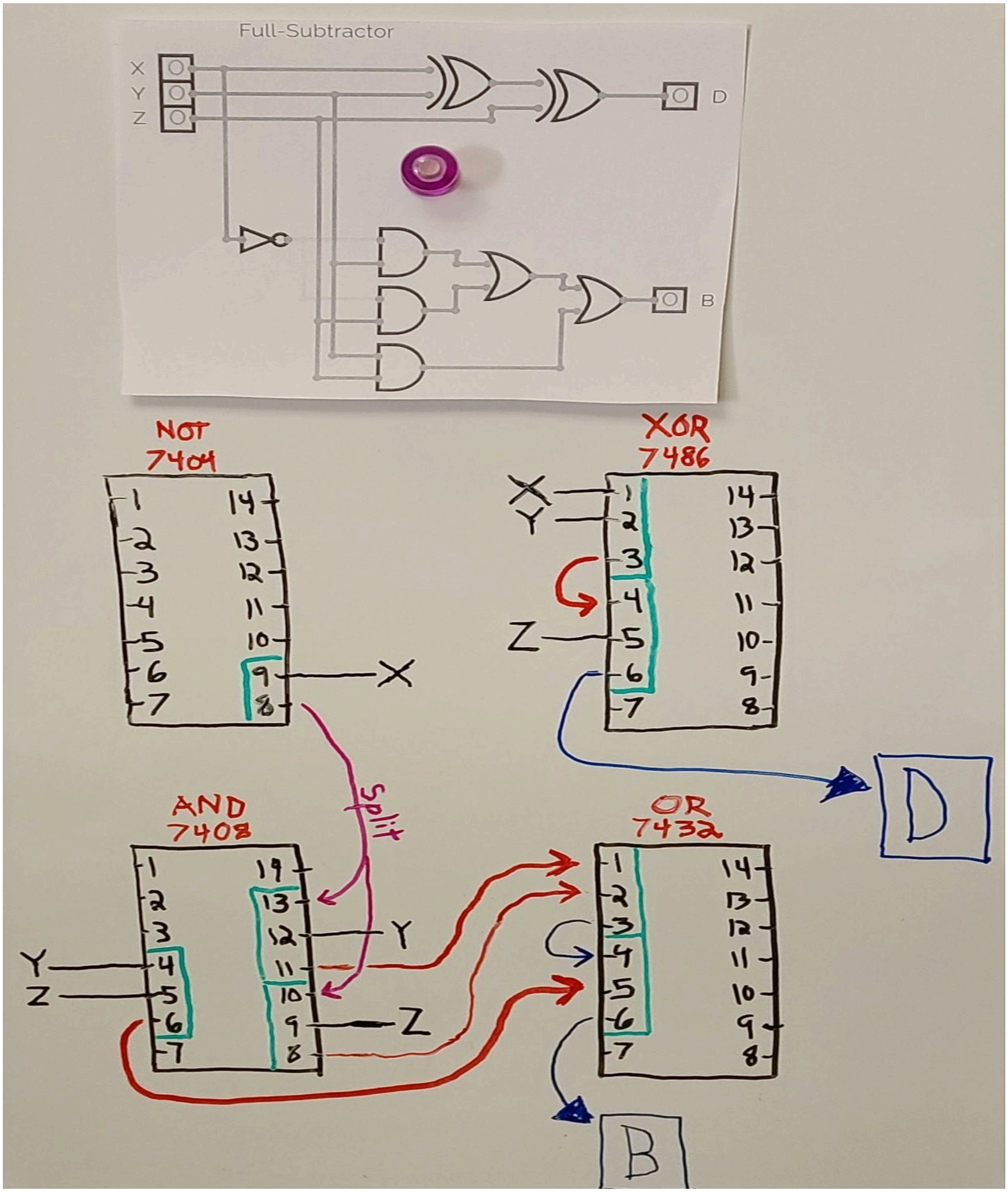
1) Truth Table



2) K-maps



3) Expression of functions D and B in terms of x,y,z
   D = X XOR Y XOR Z = XYZ + XY'Z' + X'YZ' + X'Y'Z
   B = X'Z + YZ + X'Y
4) Circuit Diagram of Full-subtractor



https://circuitverse.org/users/403924/projects/lab-5-group-2

5) Prove that functions D and B 4) are equal to 3)

$$(x \oplus y) \oplus z$$

$$x \oplus y = xy' + x'y$$

$$(xy' + x'y) \oplus z$$

$$(xy' + x'y)z' + (xy' + x'y)'z$$

$$xy'z' + x'yz'$$

$$(xy')'(x'y)'$$

de Morgan

$$(x' + y)(x + y')$$

$$\underbrace{\cancel{x'x}}_{0} + x'y' + yx + \underbrace{\cancel{yy'}}_{0}$$

$$(x'y' + yx)z$$

$$x'y'z + yxz$$

$$\boxed{xy'z' + x'yz' + x'y'z + yxz}$$

Q.E.D.

6) Observations

   Our circuit was observed to match the expected truth table values for D and B